

Software Engineering project

AI Based Inventory Management System

Synopsis

Efficient inventory management is crucial for businesses to minimize waste, optimize stock levels, and ensure smooth operations. Traditional inventory systems often rely on manual tracking or static rule-based approaches, leading to inefficiencies and stock discrepancies. This project aims to develop an AI-powered inventory management system that predicts demand, automates restocking, and prevents overstock or shortages. It will integrate real-time tracking using IoT devices and blockchain for secure supply chain transparency, benefiting businesses of all sizes.

Input

- **Stock Data:** Current inventory levels, item details, stock movement
- **Sales Data:** Purchase orders, sales transactions, seasonal demand patterns
- **Supplier Information:** Vendor details, delivery schedules, order history
- **External Data:** Market trends, competitor pricing, supply chain disruptions

Functions

- **Real-Time Inventory Tracking**
- **Automated Stock Replenishment**
- **AI-Based Demand Forecasting**
- **Supplier and Order Management**
- **Blockchain Integration for Supply Chain Transparency**

Output

- **Stock Status Reports (In-Stock, Low Stock, Out of Stock)**
- **Automated Reorder Alerts**

- Demand Forecasting Insights
- Supplier Performance Reports
- Audit Logs for Regulatory Compliance

